

Assignment 6

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1 How to prove that $3SAT \leq_P 3COLOR$ and thus $3COLOR$ is NP-complete?

We prove that $3SAT \leq_P 3COLOR$, which implies $3COLOR$ is NP-complete.

First, $3COLOR \in NP$. Given a coloring of the graph, we can check every edge and verify in polynomial time that its two endpoints receive different colors.

Now we reduce $3SAT$ to $3COLOR$. We describe the construction using the following example:

$$\phi = (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z}).$$

The construction is built from three kinds of gadgets. We first describe each one in isolation and then explain how they are wired together.

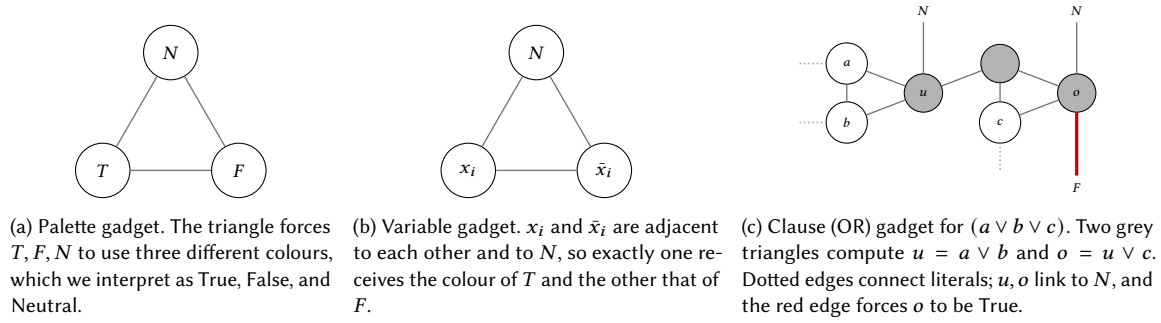


Fig. 1. The three elementary gadgets.

Palette gadget (Figure 1a). Three vertices T, F, N form a triangle. They must receive three distinct colours, which we call True, False, and Neutral. All other gadgets anchor to these three vertices.

Variable gadget (Figure 1b). For each Boolean variable x_i we create two vertices x_i and $\bar{x}_i$. Both are joined to N and to each other, forming a triangle with N . Because x_i and $\bar{x}_i$ are adjacent to N , neither can take the neutral colour; because they are adjacent to each other, exactly one receives the colour of T (and is interpreted as True) while the other receives the colour of F (and is interpreted as False). Thus the gadget forces a consistent truth assignment.

Clause gadget (Figure 1c). For each clause $(a \vee b \vee c)$ we chain two OR-gadgets:

- (1) The first grey triangle has inputs a, b and output $u = a \vee b$.
- (2) The second grey triangle has inputs u, c and output $o = u \vee c$.

The intermediate output u and the final output o are each connected to the palette vertex N , so both must be coloured either True or False. The final output o is also joined to the palette vertex F by a red edge. Since o is adjacent to F , it cannot itself be False; therefore o must be True. The internal structure of the OR-gadget guarantees that o can be True exactly when at least one of a, b, c is True. Hence the entire clause gadget is 3-colourable iff the clause is satisfied.

Wiring the gadgets together. Given a 3SAT formula ϕ , the complete graph G_ϕ is assembled as follows:

- (1) **Add the palette.** Insert the three palette vertices T, F, N and the edges among them.
- (2) **Add variable gadgets.** For each variable x_i in ϕ , create vertices x_i and $\bar{x}_i$, add the edges $(x_i, \bar{x}_i)$, (x_i, N) , $(\bar{x}_i, N)$.
- (3) **Add clause gadgets.** For each clause $C_j = (a_j \vee b_j \vee c_j)$ in ϕ , instantiate a chain of two OR-gadgets with inputs a_j, b_j, c_j , intermediate output u_j , and final output o_j as described above. Connect u_j to N , connect o_j to N , and connect o_j to F .
- (4) **Connect literals to clauses.** Each literal vertex (x_i or $\bar{x}_i$) is linked by a dotted edge to every OR-gadget input that corresponds to an occurrence of that literal in a clause. (These dotted edges are the only connections between the variable layer and the clause layer.)

The construction uses a constant number of vertices and edges per variable and per clause, so G_ϕ has size polynomial in $|\phi|$ and can be built in polynomial time.

Figure 2 shows the complete graph for the example formula $\phi = (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$.

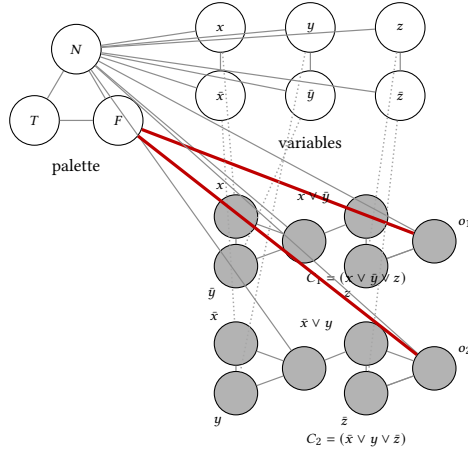


Fig. 2. Reduction from 3SAT to 3COLOR for the formula $\phi = (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee y \vee \bar{z})$. Palette vertices T, F, N enforce the three colors; variable gadgets force a consistent truth assignment; clause gadgets (gray vertices) propagate the OR of literals and force the final output to be True via the red edges.

Correctness. If ϕ is satisfiable, colour each literal vertex according to its truth value under a satisfying assignment (colour of T for True, colour of F for False). Because every clause has at least one true literal, each OR-gadget can propagate True to its final output o_j , which is compatible with the red edge to F and the edge to N . Hence G_ϕ admits a proper 3-colouring.

Conversely, suppose G_ϕ has a proper 3-colouring. The variable gadgets force each pair $(x_i, \bar{x}_i)$ to receive opposite non-neutral colours, so we read off a Boolean assignment by interpreting the colour of T as True. If a clause C_j were unsatisfied under this assignment, all three of its literal inputs would be coloured False. The OR-gadgets would then force the final output o_j to be False as well. But o_j is adjacent to F (red edge), so o_j cannot take the colour of F —a contradiction. Therefore every clause is satisfied and ϕ is satisfiable.

The reduction is polynomial-time because each variable and each clause contributes only $O(1)$ vertices and edges. Thus $3\text{SAT} \leq_P 3\text{COLOR}$. Since 3SAT is NP-complete and $3\text{COLOR} \in \text{NP}$, we conclude that 3COLOR is NP-complete.